SLIIT UNI
THE KNOWLEDGE UNIVERSITY

IT2011 - Artificial Intelligence and Machine Learning

Department of Information Technology, Faculty of Computing

Year 2 semester 1 (2026)

Tutorial 04

Supervised Learning Workshop 1

PART A — Linear Regression

Question A1

Scenario: Predicting Daily Coffee Sales from Temperature

A cafe owner believes that colder days lead to higher hot-coffee sales. She records outdoor temperature (°C) and the number of coffees sold on 5 days:

Day	Temperature x (°C)	Coffees Sold y
Mon	8	95
Tue	12	80
Wed	16	70
Thu	20	55
Fri	24	45

(a) Calculate the means $\bar{x}$ and $\bar{y}$.

(b) Complete the OLS working table and calculate β_1 and β_0 using the formulas:

$\theta_1 = \Sigma(x_i - \bar{x})(y_i - \bar{y}) \;/\; \Sigma(x_i - \bar{x})^2 \qquad \theta_0 = \bar{y} - \theta_1 \times \bar{x}$

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$

(c) Write the regression equation $\hat{y} = \beta_0 + \beta_1 x$. Then predict coffee sales on a 10 °C day.

(d) Calculate the residuals ($y_i - \hat{y}_i$) for each of the 5 days and then compute SSE = $\Sigma(y_i - \hat{y}_i)^2$.

Day	Actual y	Predicted $\hat{y}$	Residual ($y - \hat{y}$)	Residual²

Question A2

Scenario: Predicting Student Exam Scores from Study Hours

A lecturer wants to predict a student's exam score (%) from the number of hours they studied. Data from 6 students:

Student	Study Hours x	Exam Score y (%)
A	2	45
B	4	58
C	6	72
D	8	80
E	10	90
F	12	95

(a) Compute $\bar{x}$ and $\bar{y}$.

(b) Calculate β_1 and β_0 using the OLS formulas. Show the full working table.

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$

(c) A new student studied for 9 hours. Use your model to predict their score.

(d) Interpret the slope β_1 in plain language: what does it mean in this context?

(e) If a student studied 0 hours, what does the model predict? Is this realistic? Briefly explain.

PART B — Multiple Linear Regression

Key Formula: $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ | Each β_i = change in $\hat{y}$ for a 1-unit increase in x_i, holding all other predictors constant.

Question B1

Scenario: Predicting House Rent from Area and Distance to City Centre

A property analyst wants to predict monthly rent (LKR 000s) using two features: floor area (m^2) and distance to city center (km). She has collected data from 5 apartments:

Apt	Area x_1 (m^2)	Distance x_2 (km)	Rent y (LKR 000s)
A	40	2	45
B	60	5	52
C	80	3	68
D	100	8	70
E	120	1	95

After solving the normal equations $X'X\beta = X'y$, the analyst finds:

$\beta_0 = 10.0$ $\beta_1 = 0.65$ $\beta_2 = -1.80$

(a) Write out the full regression equation $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ with the given values substituted in.

(b) Use the equation to predict the monthly rent for an apartment with floor area = 90 m^2 and distance = 4 km. Show all working.

(c) Complete the table of predicted values and residuals for all 5 apartments.

Apt	x_1	x_2	Actual y	Predicted $\hat{y}$	Residual $(y - \hat{y})$

(d) Interpret $\beta_1 = 0.65$ and $\beta_2 = -1.80$ in plain language. What does the negative sign on β_2 tell us?

(e) A second analyst suggests using only x_1 (area) and dropping x_2 (distance). Give one reason why keeping both predictors might produce better predictions.

Question B2

Scenario: Predicting Paddy Crop Yield from Rainfall and Fertilizer

An agricultural researcher models paddy yield (tones/hectare) using two predictors: monthly rainfall (mm) and fertilizer applied (kg/hectare). Six plots of land were observed during the same growing season:

Plot	Rainfall x_1 (mm)	Fertiliser x_2 (kg/ha)	Yield y (t/ha)
1	150	50	3.2
2	180	80	4.1
3	200	100	5.0
4	220	60	4.8
5	170	120	4.5
6	250	90	6.0

The fitted model (obtained from the normal equations) is:

$$\hat{y} = -0.50 + 0.022\, x_1 + 0.018\, x_2$$

(a) Predict the yield for a plot receiving 190 mm rainfall and 70 kg/ha fertiliser. Show your calculation.

(b) Calculate the predicted value $\hat{y}$ and residual $(y - \hat{y})$ for each of the 6 plots.

Plot	x_1	x_2	Actual y	Predicted $\hat{y}$	Residual $(y - \hat{y})$	Residual^2

(c) Sum the residuals. What do you notice? Why does this always happen with OLS regression?

(d) Compute SSE = $\Sigma(y_i - \hat{y}_i)^2$ using your table above.

(e) Interpret $\beta_1 = 0.022$ in context. Then explain why we cannot simply compare β_1 and β_2 to decide which predictor is 'more important'.

PART C — Logistic Regression

Key Formulas to Use | $z = \beta_0 + \beta_1 x$ $\sigma(z) = 1 / (1 + e^{-z})$ Decision: $P \geq 0.5 \rightarrow$ Class 1 ; $P < 0.5 \rightarrow$ Class 0

Question C1

Scenario: Predicting Diabetes Risk from Blood Sugar Level

A hospital clinician uses blood sugar level (mmol/L) to screen patients for diabetes risk. A logistic regression model has been trained with parameters:

$\beta_0 = -10$ $\beta_1 = 1.5$

(a) For each patient below, compute $z = \beta_0 + \beta_1 x$, then $P = \sigma(z)$, then state the predicted class.

Patient	Blood Sugar x (mmol/L)	$z = \beta_0 + \beta_1 x$	$P = \sigma(z)$	Predicted Class
P1	4			
P2	6			
P3	7			
P4	8			
P5	10			

(b) At what value of x does the decision boundary occur (i.e., where z = 0)? Show your algebra.

(c) The true labels for P1–P5 are: 0, 0, 1, 1, 1. List any misclassified patients and explain what the model got wrong.

(d) Why can't we use linear regression to make this binary prediction? Give one concrete reason.

Question C2

Scenario: Spam Email Detection Using Word Count

An email client uses the number of suspicious words in an email to classify it as Spam (1) or Not Spam (0). The trained model parameters are:

$\beta_0 = -5$ $\beta_1 = 1.2$

(a) Complete the prediction table. For each email, calculate z, P, and the predicted label.

Email	Suspicious Words x	$z = \beta_0 + \beta_1 x$	$P = \sigma(z)$	Predicted Class
E1	2			
E2	4			
E3	5			
E4	6			
E5	8			

(b) Solve for the decision boundary x* where P = 0.5 exactly. Show all algebraic steps.

(c) True labels: E1=0, E2=0, E3=1, E4=1, E5=1. Calculate the model's classification accuracy on these 5 emails.

(d) If the threshold is changed from 0.5 to 0.3, how does this affect: (i) which emails are classified as spam, and (ii) the trade-off between false positives and false negatives?

Linear Regression

Consider a subset of the Boston Housing Dataset, a well-known real-world dataset used to predict median house prices.

Sample	RM (x_1)	LSTAT (x_2)	PTRATIO (x_3)	MEDV (y)
1	6.575	4.98	15.3	24.0
2	6.421	9.14	17.8	21.6
3	7.185	4.03	17.8	34.7
4	6.998	2.94	18.7	33.4
5	7.147	5.33	18.7	36.2

Where:

- x_1= Average number of rooms per dwelling (RM)
- x_2= % lower status population (LSTAT)
- x_3= Pupil-teacher ratio by town (PTRATIO)
- y= Median value of owner-occupied homes in $1000s (MEDV)

Assume that the relationship between housing features and price is modelled using multiple linear regression:

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

Use the following parameter values:

$$\theta_0 = -5.0, \theta_1 = 4.5, \theta_2 = -0.6, \theta_3 = -0.3$$

1. Compute the predicted house price for each sample.
2. Calculate the Mean Squared Error (MSE).
3. Calculate the coefficient of determination (R^2).
4. What does a low MSE indicate?
5. What does an R^2 value close to 1 mean?
6. Which feature seems most influential in predicting house prices?

Logistic Regression

Consider a subset of the Pima Indians Diabetes Database, a well-known real-world medical dataset used for predicting the onset of diabetes.

The dataset contains medical measurements of female patients of Pima Indian heritage. The target variable:

- $y = 1 \rightarrow$ Diabetes
- $y = 0 \rightarrow$ No Diabetes

Below is a small subset with five samples:

Sample	Pregnancies (x_1)	Glucose (x_2)	BMI (x_3)	(y)
1	2	138	33.6	1
2	1	85	26.6	0
3	8	183	23.3	1
4	1	89	28.1	0
5	5	116	25.6	1

Assume the probability of diabetes is modeled using logistic regression:

$$z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$
$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Use the following parameter values:

$$\theta_0 = -8.0, \theta_1 = 0.15, \theta_2 = 0.04, \theta_3 = 0.10$$

1. Compute the predicted probability for each sample.
2. Using a classification threshold of 0.5, determine the predicted class labels.

K-Nearest Neighbor

Consider the following dataset with eight training samples for a loan approval classification problem.

Applicant	Income (₹1000s)	Credit Score	Loan Amount (₹1000s)	Loan Status
1	40	650	200	Approved
2	25	600	150	Rejected
3	60	720	250	Approved
4	35	580	120	Rejected
5	50	690	220	Approved
6	28	610	130	Rejected
7	55	710	240	Approved
8	30	590	140	Rejected

1. Calculate the means and standard deviations for each numerical column.

2. Create the scaled dataset using Standardization method for the numerical columns. (Note: ($Z = \frac{x-\mu}{\sigma}$))

3. Using the above training set, determine the K-Nearest Neighbor classification for the following test applicant:

Income (₹1000s)	Credit Score	Loan Amount (₹1000s)
45	670	210

(Note: Use Manhattan distance as the distance measure, i.e $d(p, q) = \sum_{i=1}^{n} |p_i - q_i|$ and K as 3)

Reflection Questions

1. What are the main differences between regression and classification?

2. When would you prefer KNN over logistic regression?

3. What happens if K is too large or too small?